Student Engagement Across Learning Activities

Study Details

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Survey and Interpretation

The one-time in-class survey includes 43 students in 3 discussion sections (March 29-30, 2017). Students rated small-group discussion, big-group discussion, a debate on globalization, and a campus field trip on a 1-5 scale for interest, informativeness, and encouragement of critical thinking. Responses marked "Didn't attend" are excluded from comparisons involving that activity. Each comparison uses students with both required ratings.

The figures show individual ratings and descriptive associations. Equal original ratings share one color, and other colors identify the higher-rated activity or dimension. Small random offsets reveal overlapping integer responses. Correlations use the original scores. The figures omit correlation intervals, which are reported in the supporting tables.

Patterns Across Activities

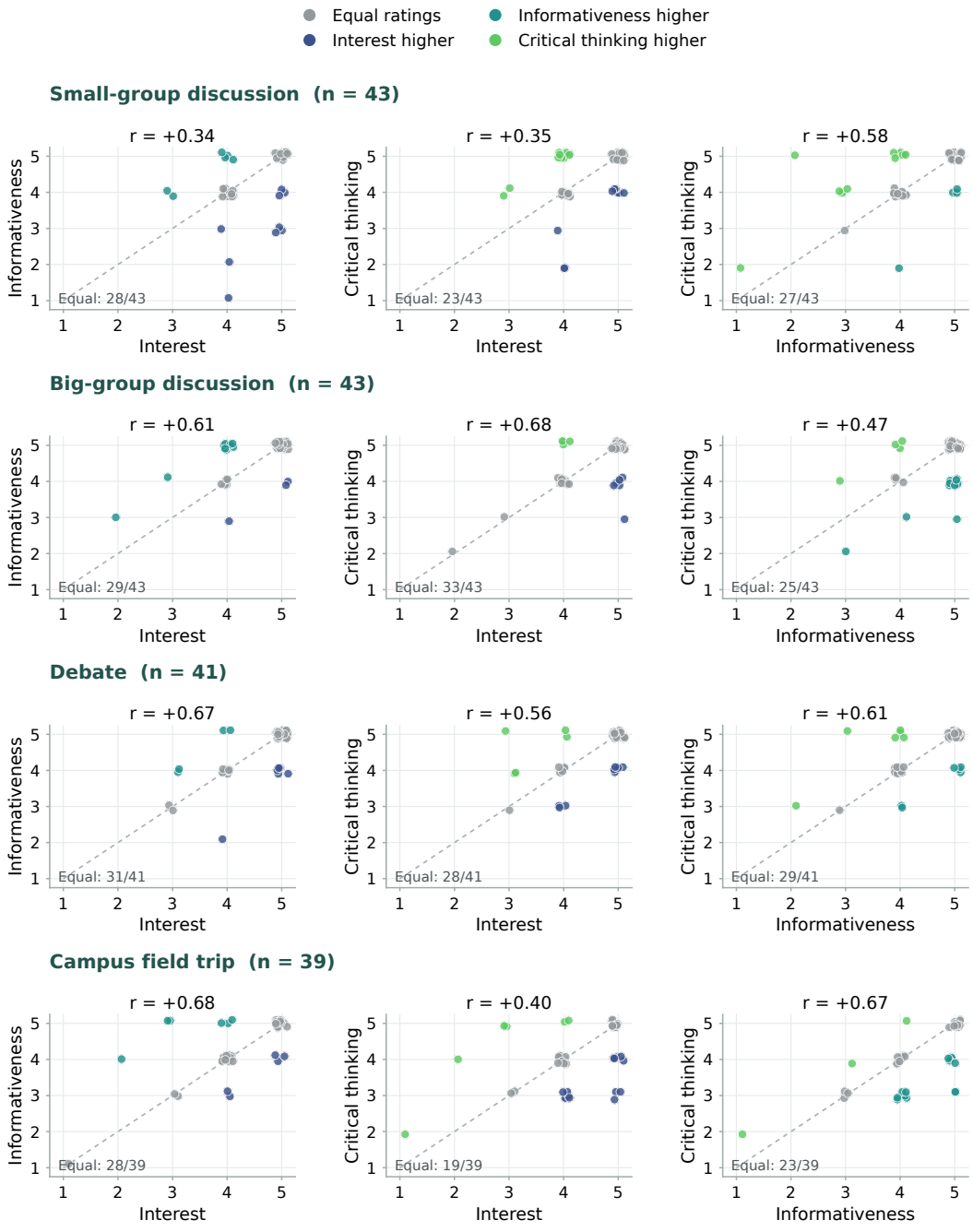
- Students who rated an activity as more interesting also tended to rate it as more informative and more encouraging of critical thinking. Within-activity correlations range from +0.34 to +0.68
- Interest in one activity was a weak guide to interest in another. The interest correlation for small- and big-group discussion is +0.11. For debate and field trip it is +0.003
- The debate and field-trip ratings are more closely associated for critical thinking, with a correlation of +0.55. These comparisons describe point estimates and do not establish differences in effects on learning

Table 1. Correlations Across Activities

Activity Pair	Interest	Informativeness	Critical Thinking
Small-group + Big-group ($n = 43$)	+0.11 [-0.187, +0.351]	-0.11 [-0.344, +0.229]	-0.21 [-0.447, +0.052]
Small-group + Debate ($n = 41$)	+0.21 [-0.060, +0.488]	+0.14 [-0.135, +0.390]	+0.04 [-0.214, +0.374]
Small-group + Field trip ($n = 39$)	+0.32 [-0.104, +0.641]	+0.10 [-0.147, +0.345]	+0.16 [-0.088, +0.429]
Big-group + Debate ($n = 41$)	+0.31 [-0.026, +0.569]	+0.28 [+0.043, +0.585]	+0.46 [+0.059, +0.736]
Big-group + Field trip ($n = 39$)	+0.24 [+0.008, +0.484]	+0.13 [-0.081, +0.416]	+0.36 [+0.123, +0.625]
Debate + Field trip ($n = 37$)	+0.00 [-0.234, +0.320]	+0.21 [-0.034, +0.539]	+0.55 [+0.324, +0.729]

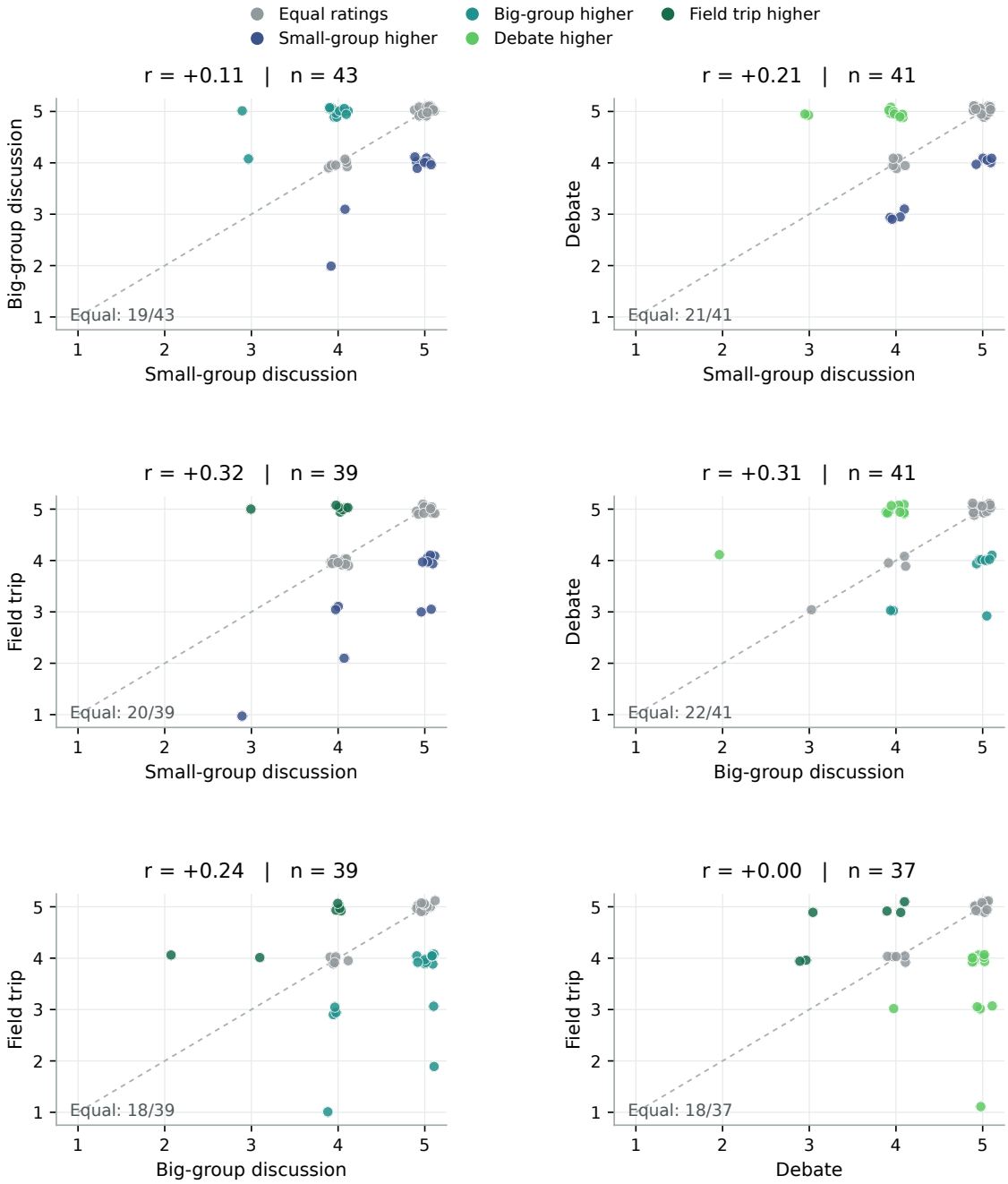
Notes. Pearson correlations. Brackets show 95% percentile intervals from 10,000 bootstrap draws of students, separately for each activity pair and dimension. Each comparison uses the same students for its two activities. Intervals are pointwise and do not adjust for the exploratory comparisons. Zero is the benchmark for a lack of linear association. Source: original GEOG 122 survey responses.

Figure 1. Ratings of Different Dimensions Within Each Activity



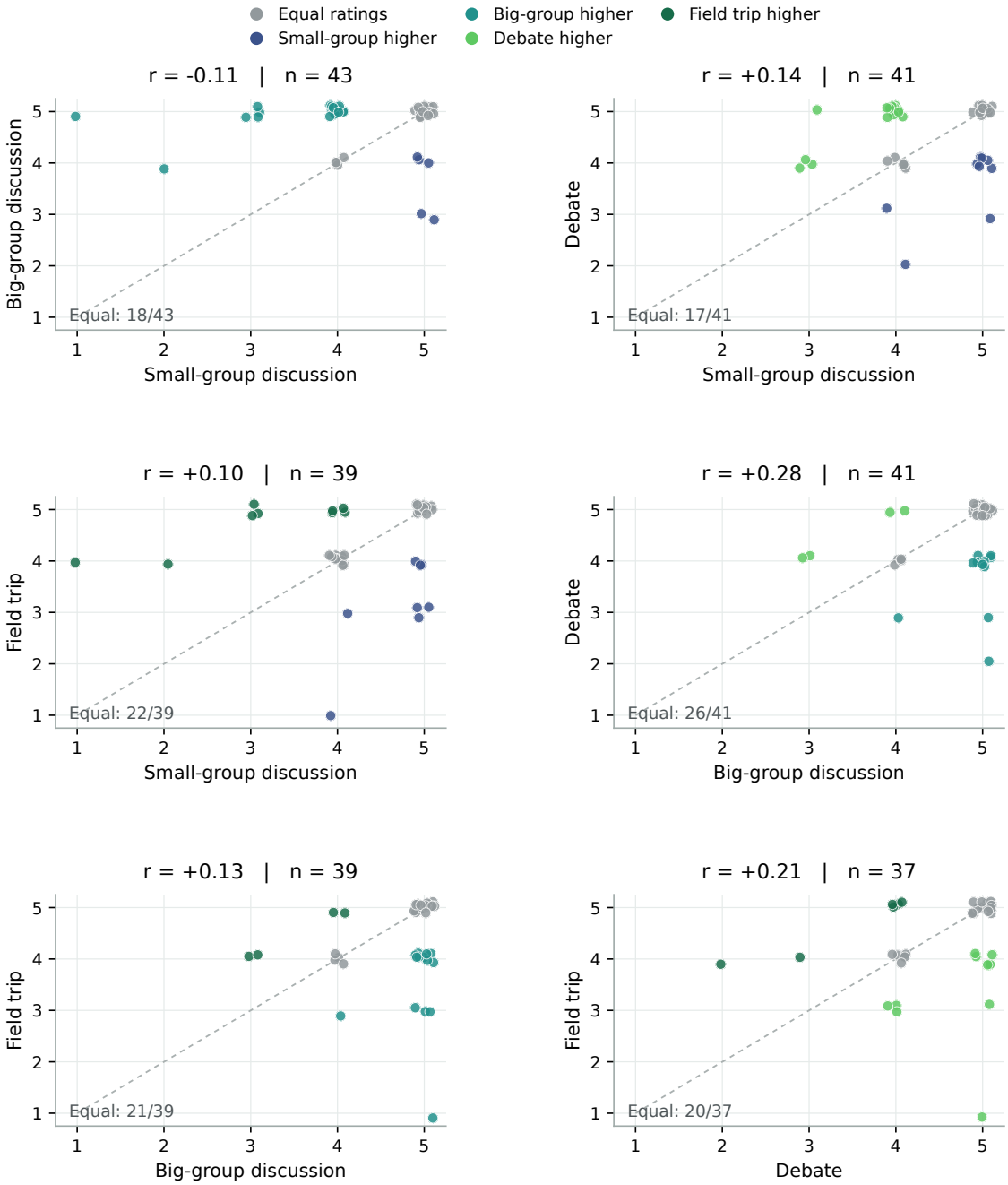
Notes. Each dot is one student's pair of ratings. Colors identify equal original scores or the dimension with the higher score. The dashed line marks equality. Small offsets reveal overlapping responses. Each activity uses students with all three ratings observed. Pearson correlations (r) use original scores. Source: GEOG 122 survey, UBC, 2017.

Figure 2. Interest Ratings Across Activities



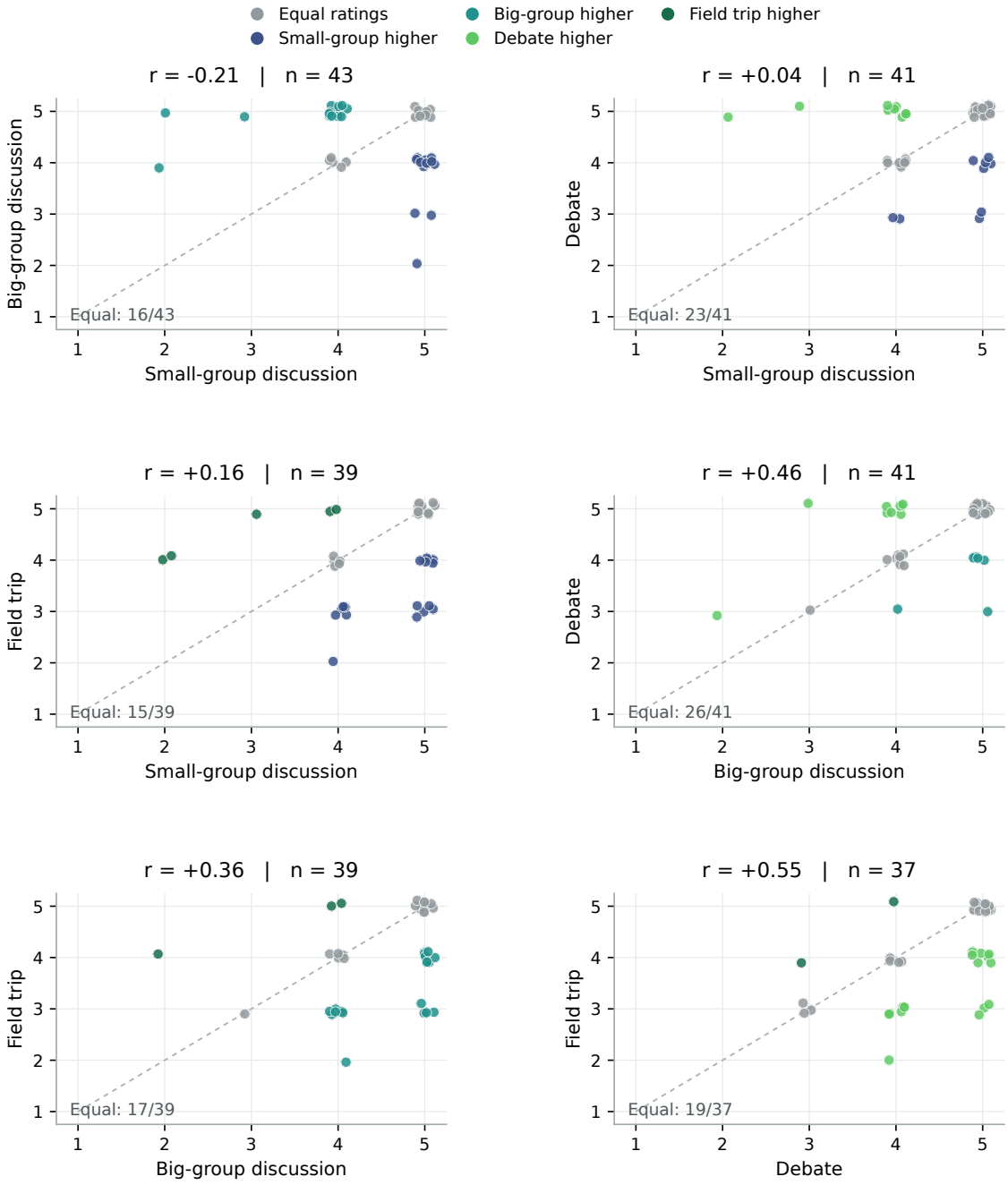
Notes. Each dot compares the same student's interest ratings for two activities. Colors identify equal scores or the activity with the higher original rating. The dashed line marks equality. Display offsets reveal overlap. Each panel includes students who rated both activities. Pearson correlations use original scores. Source: GEOG 122 survey, UBC, 2017.

Figure 3. Informativeness Ratings Across Activities



Notes. Each dot compares the same student's informativeness ratings for two activities. Colors identify equal scores or the activity with the higher original rating. The dashed line marks equality. Display offsets reveal overlap. Each panel includes students who rated both activities. Pearson correlations use original scores. Source: GEOG 122 survey, UBC, 2017.

Figure 4. Critical-Thinking Ratings Across Activities



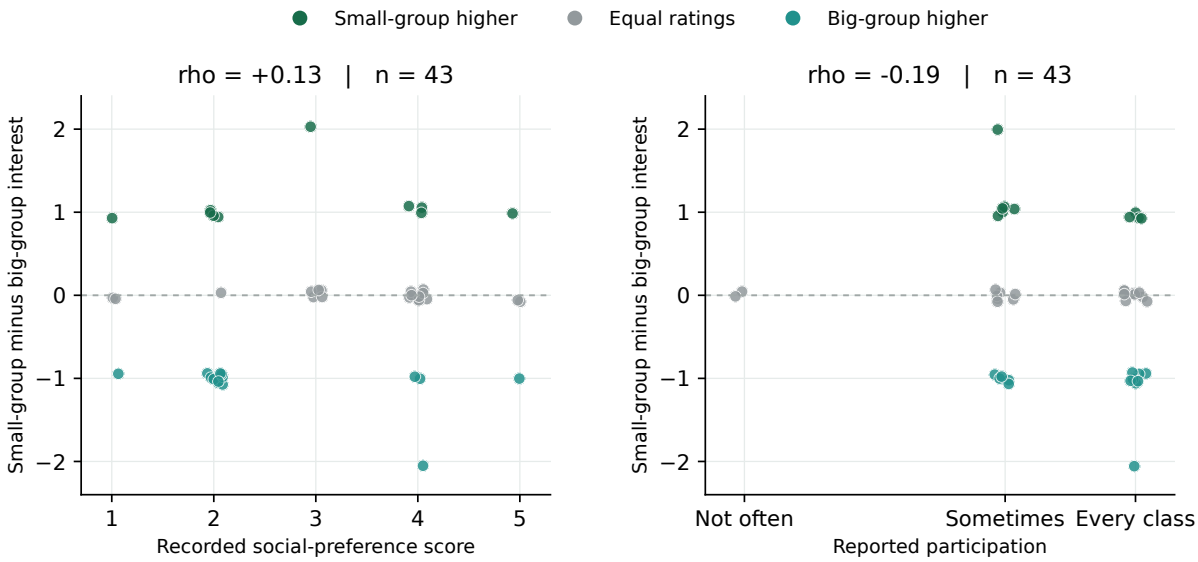
Notes. Each dot compares the same student's ratings of how much two activities encouraged critical thinking. Colors identify equal scores or the activity with the higher original rating. The dashed line marks equality. Display offsets reveal overlap. Each panel includes students who rated both activities. Pearson correlations use original scores. Source: GEOG 122 survey, UBC, 2017.

Social Preferences and Participation

The survey also asks "Are you more of an extrovert or introvert?" and records scores from 1 to 5. The endpoint labels are unavailable in the saved sources, so the direction cannot be interpreted as more introverted or more extroverted. The analysis retains the original numeric score. Participation responses are labeled "Not often," "Sometimes," and "In every class."

Among 43 students, interest in small-group discussion minus interest in big-group discussion has a rank correlation of +0.13 with the recorded social score and -0.19 with participation. These associations are small and imprecise.

Figure 5. Social Responses and Relative Interest in Discussion Formats



Notes. Positive vertical values indicate a higher interest rating for small-group discussion, negative values indicate a higher big-group rating, and zero indicates a tie. Each dot is one student. Colors use original scores. Small display offsets reveal overlapping values. The reported statistic is Spearman's rank correlation. The social-score endpoints remain unassigned. Source: original GEOG 122 survey responses.

The broader exploratory checks in Table 2 do not establish clear associations. All bootstrap intervals include zero. Permutation tests adjusted across the 13 comparisons with Holm's method also remain above the 5% threshold. These responses do not establish stable student types or causal effects of activities on learning.

Table 2. Exploratory Associations With Social Responses

Rating or Contrast	Predictor	N	Rank Corr.	95% Interval
Small minus big interest	Social score	43	+0.13	[-0.17, +0.44]
Small minus big interest	Participation	43	-0.19	[-0.45, +0.10]
Debate minus field interest	Social score	37	+0.12	[-0.22, +0.44]
Debate minus field interest	Participation	37	-0.08	[-0.42, +0.27]
Small-group interest	Social score	43	+0.09	[-0.20, +0.40]
Small-group interest	Participation	43	+0.02	[-0.28, +0.32]
Big-group interest	Social score	43	-0.08	[-0.37, +0.22]
Big-group interest	Participation	43	+0.26	[-0.04, +0.52]
Debate interest	Social score	41	-0.00	[-0.29, +0.30]
Debate interest	Participation	41	+0.16	[-0.17, +0.46]
Field trip interest	Social score	39	-0.19	[-0.51, +0.16]
Field trip interest	Participation	39	+0.21	[-0.10, +0.50]
Participation	Social score	43	-0.20	[-0.48, +0.09]

Notes. Spearman rank correlations with 95% percentile intervals from 10,000 bootstrap draws of students. Ranks are recomputed within each draw. Each comparison uses observed pairs. Participation preserves the ordering of the labeled response categories. The social score retains its original numeric coding because endpoint labels are unavailable. The contrasts subtract the second activity's interest rating from the first. The intervals are pointwise. Zero is the benchmark for a lack of rank association. Source: original GEOG 122 survey responses.

Table 3. Correlations Within Each Activity

Activity	Students	Interest + informativeness	Interest + critical thinking	Informativeness + critical thinking
Small group discussion	43	+0.34 [+0.117, +0.558]	+0.35 [+0.147, +0.525]	+0.58 [+0.211, +0.829]
Big group discussion	43	+0.61 [+0.249, +0.820]	+0.68 [+0.257, +0.886]	+0.47 [+0.042, +0.737]
Debate	41	+0.67 [+0.487, +0.852]	+0.56 [+0.305, +0.783]	+0.61 [+0.298, +0.822]
Campus field trip	39	+0.68 [+0.303, +0.880]	+0.40 [+0.004, +0.677]	+0.67 [+0.440, +0.828]

Notes. Pearson correlations. Brackets show 95% percentile intervals from 10,000 student bootstrap draws within each activity. Each row uses students with all three ratings observed. "Didn't attend" answers are excluded. Intervals are pointwise. Zero is the benchmark for a lack of linear association. Source: original GEOG 122 survey responses.

Source and Scope

Original GEOG 122 learning-activity feedback survey, University of British Columbia, 2017. The same respondents contribute to several comparisons. Missing activity ratings reflect nonattendance and are not filled. Results describe a small course sample, with many tied ratings and scores near the upper limit of the scale.